ArthoCare

Project Team

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Session 2022-2026

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June, 2026

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Chapter 1

Introduction

The purpose of this document is to propose the development of ArthoCare, a digital health system designed to provide personalized support for patients with Rheumatoid Arthritis (RA). The system integrates computer vision, multimodal machine learning, lifestyle monitoring, and weather-based predictions into a multi-agent architecture. By combining these modalities, ArthoCare aims to deliver a personalized early warning system for RA flares.

In addition to empowering patients with tailored exercise and lifestyle recommendations, the project enables patients to generate summarized reports that can be shared with clinicians during in-person visits. Since RA is a chronic autoimmune disease that progressively impacts joints and mobility, early identification of flares and personalized interventions are critical *(Smolen et al., 2020)*. ArthoCare leverages recent advances in digital health research and mobile technologies to bridge this clinical gap.

1.1 Problem Statement

Patients with RA often experience unpredictable flare-ups, characterized by pain, stiffness, and inflammation *(Smolen et al., 2020)*. These flares cause reduced mobility, lost productivity, and high healthcare costs *(Cross et al., 2014)*. Current management depends heavily on infrequent in-person visits, which hinders continuous monitoring and predictive care.

Existing mobile health tools mostly focus on generic symptom logging or exercise recommendations, but they lack multimodal integration (weather, lifestyle, biomechanics) and transparent predictive modeling. Recent studies confirm the importance of these factors: lifestyle and demographic data improve RA outcome prediction *(O'Neill et al., 2022)*, weather factors strongly influence pain (Druce et al., 2019; Wadström et al., 2024), and computer vision can reliably track range of motion for musculoskeletal disorders *(Fer-*

reira et al., 2021). Therefore, there is a pressing need for a digital health platform that combines these heterogeneous data sources into a predictive and personalized system, empowering patients to manage their condition and to share structured reports with their clinicians.

1.2 Motivation

The motivation for *ArthoCare* is threefold:

1. **Clinical Need:** Rheumatoid Arthritis (RA) affects over 1% of the global population *(cross et al., 2014)*, with rising prevalence in South Asia, including Pakistan. (farooqi Gibson, 1998). Flare prediction and self-management are vital for reducing disease burden*(Smolen et al., 2020).*

2. **Scientific Evidence:** Studies such as *Cloudy with a Chance of Pain* (2019) and *Wadström et al. (2024)* have shown links between weather conditions and RA pain. Lifestyle factors such as stress, BMI, and sleep significantly influence outcomes (O'Neill et al., 2022).Research also highlights the value of lifestyle and demographic data in predicting RA progression. Furthermore, advances in computer vision for range-of-motion (ROM) assessment *(Ferreira et al., 2021)* confirm the feasibility of digital biomechanics monitoring.

3. **Technological readiness:** Modern smartphones can perform on-device computer vision for joint analysis, while APIs provide real-time weather data. Privacy-preserving AI ensures these tools are deployed securely and cost-effectively *(Topol, 2019).*

ArthoCare addresses both the scientific opportunity and the clinical gap, enabling personalized RA management in a resource-efficient manner. .

1.3 Problem Solution

ArthoCare employs a multi-agent architecture, where each agent specializes in a task such as monitoring lifestyle, integrating weather, biomechanics analysis, prediction, or recommendations. Together, they form a unified early-warning system.

The system aims to:

1. **Predict who** is most likely to experience a flare (Lifestyle & Demographics), validated by studies on RA progression *(O'Neill et al., 2022).*

2. **Predict when** flares are more likely to occur (Weather factors).
3. **Predict where** flares may occur in the body (*RALens* – ROM & biomechanics).
4. **Deliver** personalized recommendations for prevention and management.
5. **Provide** patients with a dashboard of progress
6. **Ensure** compliance with data privacy standards (HIPAA equivalent, PDPA 2023 in Pakistan).

This modular approach ensures scalability, adaptability, and clinical utility.

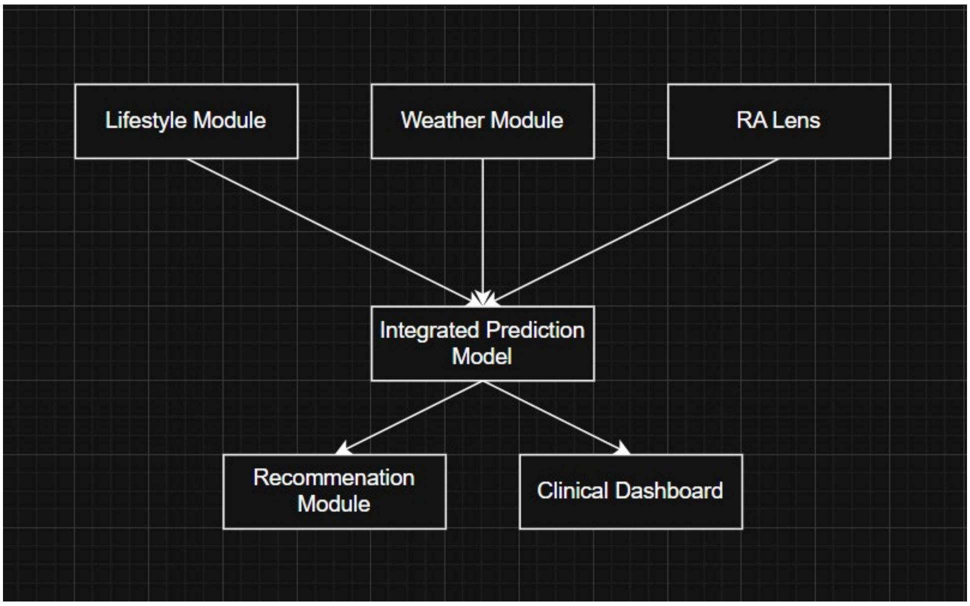

Figure 1.1: Proposed ArthoCare System Architecture

1.4 Stakeholders

- **Patients with Rheumatoid Arthritis:** Primary users who benefit from early warning and exercise recommendations.
- **External Collaborators (GIMES collaborators):** provide clinical validation, domain expertise, and support for the integration of evidence-based practices into the system.
- **Healthcare Institutions:** Potential adopters of the platform.
- **Developers/Research Team:** Responsible for design, development, and deployment.

- **Regulatory Authorities:** Ensure compliance with local data privacy laws.

Chapter 2

Project Description

2.1 Scope

ArthoCare is envisioned as a patient-centric, multi-agent mobile application designed to support individuals with Rheumatoid Arthritis (RA) through continuous self-monitoring, flare prediction, and evidence-based lifestyle guidance. The system integrates biomechanics assessment (via smartphone-based pose estimation), environmental weather conditions into a predictive framework that generates personalized flare risk assessments. At its core, the system incorporates a Lifestyle and Demographic Module that records patient information such as age, gender, BMI, activity level, sleep patterns, stress, and medication history, enabling the identification of risk factors linked to rheumatoid arthritis (RA) flares. Complementing this, the Weather Module integrates real-time local environmental variables, including temperature, humidity, and barometric pressure, which are recognized as significant contributors to symptom variability. To capture functional health, the RALens Biomechanics Module leverages smartphone cameras and pose estimation techniques to assess the range of motion across key joints—such as the hand, wrist, elbow, knee, and ankle—providing early detection of reduced mobility. These heterogeneous data streams converge in the Prediction Model, which applies multimodal learning to generate individualized flare risk assessments. To support patient engagement, the Data Logging Module facilitates the daily recording of self-reported outcomes such as pain, stiffness, fatigue, mood, and activity levels, thereby enriching the dataset and promoting self-awareness. Building on this, the Recommendation Module delivers AI-personalized, clinician-validated exercise and lifestyle interventions tailored to each patient's condition. Finally, the Privacy and Security Module ensures that all collected data are managed under HIPAA-equivalent standards, with specific alignment to Pakistan's PDPA 2023 draft, thereby safeguarding patient confidentiality. Through this modular yet integrated design, ArthoCare establishes a scalable, adaptable, and clinically relevant solution for improving RA self-management and healthcare delivery.

For development, the Lifestyle Module will utilize public health datasets such as NHANES to establish baseline risk factors, the Weather Module will combine prior research findings with real-time API inputs, and the RALens Module will be implemented using pose-estimation frameworks and validated on 20 GIMES patients. The integrated Prediction Model will unify these multimodal inputs in a proof-of-concept system to evaluate feasibility and accuracy.

2.2 Modules

The proposed system *ArthoCare* is composed of multiple modules that together enable patient-centric rheumatoid arthritis management. The modules are logically grouped into three categories: Client Mobile App for patients, and Core System Modules that support prediction and data security at the backend.

2.2.1 Client Mobile App

2.2.1.1 Lifestyle & Demographic Module

This module records essential patient information and lifestyle variables to establish a personalized health profile.

1. Records age, gender, BMI, comorbidities, and medication history.
2. Monitors lifestyle factors such as sleep, activity, stress, and fatigue.
3. Smoking and alcohol consumption.

Data Source/Approach: Baseline risk models will be trained using publicly available datasets such as NHANES and validated with patient-reported data from GIMES.

2.2.1.2 Weather Module

This module integrates local weather data as predictive features for flare occurrence.

1. Temperature (daily average and variation)
2. Tracks humidity, precipitation, wind speed, and barometric pressure.

Data Source/Approach: Real-time environmental variables will be obtained via trusted weather APIs, informed by prior studies such as Cloudy with a Chance of Pain (Druce et al., 2019) and Wadström et al. (2024).

2.2.1.3 RALens (Biomechanics Module)

This module leverages the smartphone camera and pose estimation techniques to assess joint mobility.

1. Uses OpenCV and MediaPipe for range-of-motion (ROM) detection.

2. **Upper Limb:** Hand (grip/open), wrist (flexion/extension), elbow (flexion/extension), shoulder (abduction/flexion).

3. **Lower Limb:** knee (flexion/extension), ankle (dorsiflexion/plantarflexion).

   Measurements include flexion/extension and mobility patterns to detect stiffness and progression.

Data Source/Approach: Developed using pose-estimation frameworks (OpenCV, MediaPipe) and validated on smartphone video recordings of 50 RA patients recruited through GIMES.

2.2.1.4 Data Logging Module

1. Daily patient-reported outcomes: pain (VAS scale), stiffness, fatigue, mood.

2. Optional wearable integration (steps, heart rate, sleep).

2.2.1.5 Recommendation Module

This module delivers lifestyle and exercise recommendations tailored to patient needs.

1. Provides AI-personalized, clinician-approved exercise routines.

2. Suggests preventive lifestyle strategies.

3. Sends reminders and motivational notifications.

2.2.1.6 Progress Dashboard

The dashboard effectively visualizes patient progress and manages care plans.

1. Graphs and trend visualization of symptoms, mobility, and flare risks.

2. Exportable summary reports for clinicians.

2.2.2 Core System Modules (Backend)

2.2.2.1 Prediction Module

This module integrates multimodal data sources to predict flare risks.

1. Multimodal flare prediction using weather, lifestyle, and biomechanics data.
2. Adaptive learning to individual patient baselines.

Data Source/Approach: Combines outputs from the Lifestyle, Weather, and RALens modules to form a proof-of-concept multimodal prediction system, tested on both public datasets and GIMES validation data.

2.2.2.2 Security & Privacy Module

This module ensures robust data protection and compliance with standards.

1. Applies AES-256 encryption and anonymization.
2. Enforces role-based access control.
3. Ensures compliance with HIPAA-equivalent standards and PDPA Pakistan 2023.

2.2.3 Multi-Agent System Approach

ArthoCare adopts a multi-agent architecture, where each functional module is implemented as a semi-autonomous agent that interacts with others to ensure scalability and modularity. Each agent is responsible for a specific task, and together they form the complete ecosystem.

1. **Patient Interaction Agent:** Handles onboarding, surveys, and user engagement.
2. **Biomechanics Agent (RALens):** Performs range-of-motion (ROM) detection and gait/movement analysis.
3. **Lifestyle & Routine Agent:** Monitors sleep, activity, fatigue, and other daily routines.
4. **Weather Agent:** Integrates API-based weather data such as temperature, humidity, and air pressure.

5. **Data Integration & Cleaning Agent:** Organizes patient data, handles missing values, and ensures data consistency.

6. **Prediction Agents:** Generate flare predictions, adapt to individual baselines, and provide uncertainty estimation.

7. **Recommendation Agent:** Converts predictions into actionable, clinician-approved recommendations.

8. **Security & Compliance Agents:** Ensure encryption, anonymization, and role-based access control.

9. **Meta-Agent(Coordinator):** Oversees communication between all agents, that is, resolves conflicts, prioritizes tasks (e.g. emergency alerts over daily surveys).

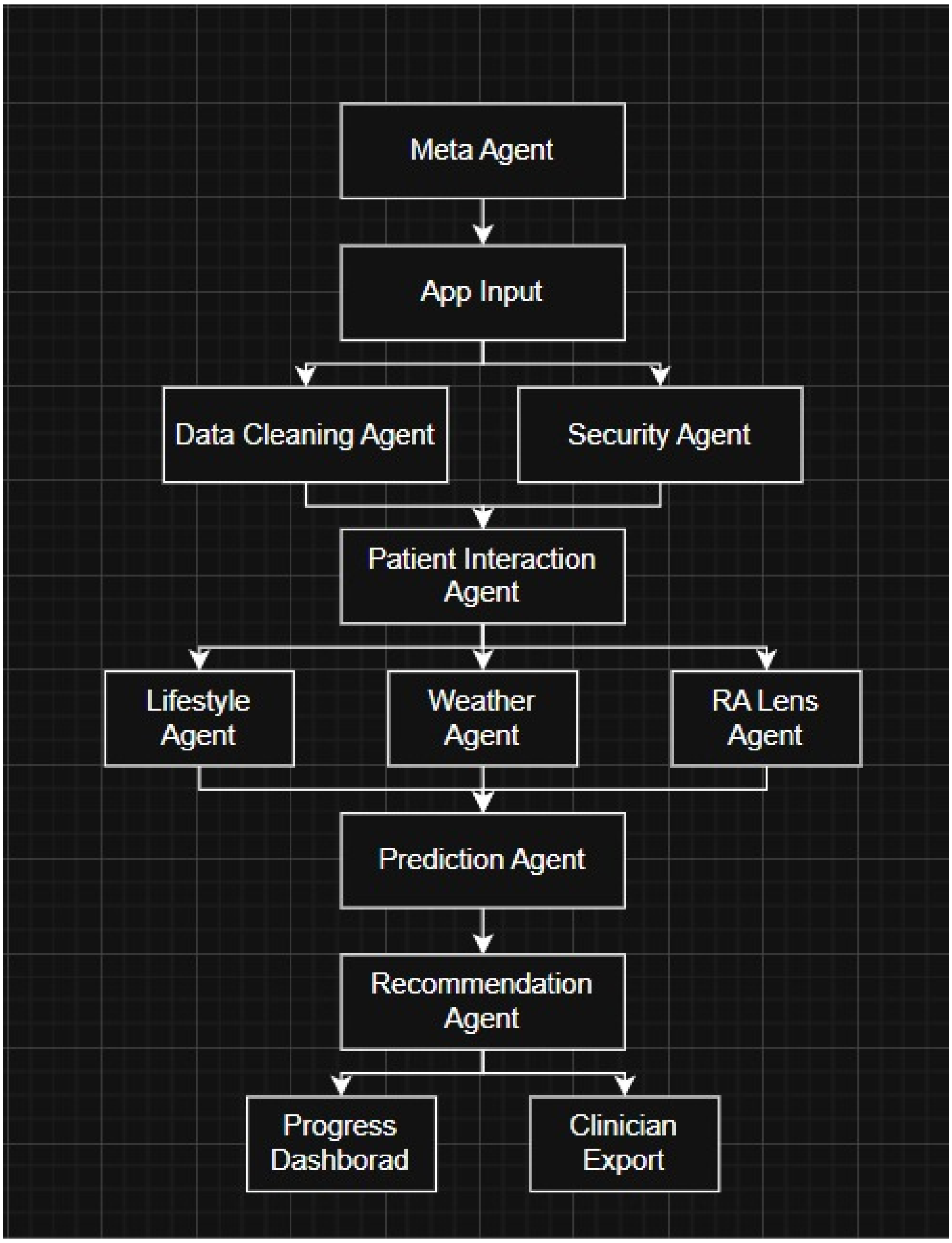

Figure 2.1: Proposed Meta-Agent Architecture

2.3 Tools and Technologies

- **Mobile App:** Flutter, Dart, SQLite/Hive.
- **Web App:** React.js, Node.js/Express, PostgreSQL.
- **ML & CV:** Python, TensorFlow Lite, PyTorch Mobile, OpenCV, MediaPipe.
- **Weather Data:** OpenWeatherMap API.

- **Security:** AES-256 encryption, JWT, OAuth2.
- **Compliance:** HIPAA/GDPR principles, Pakistan's PDPA-2023 draft.
- **Collaboration:** GitHub, Jira, Slack.

2.4 Work Division

Table 2.1: Work Division Among Team Members

Name	Registration	Responsibility / Module / Feature
Ms. Umema Ashar	22i-2036	Lifestyle + Weather Modules, Database design, Admin Module
Mr. Raja Hamza Asad	22i-1908	Mobile App (frontend), Patient Data Logging, Progress Dashboard, Exercise Module
Ms. Heer	22i-2371	RALens (CV), Flare Prediction Model, Clinician Dashboard, Security compliance

2.5 Timeline

Table 2.2: Project Timeline

Iteration #	Time Frame	Tasks / Modules
01	Sept – Oct	Setup, Authentication, Database, Lifestyle Module, Weather API integration
02	Nov – Dec	RALens prototype (hand/wrist), Patient Data Logging, Clinician Dashboard (basic)
03	Jan – Feb	RALens expansion (elbow/knee), Progress Dashboard, Flare Prediction integration, Notifications
04	Mar – Apr	Exercise Recommendation Module, Admin & Security, Full integration & pilot testing at GIMES

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